

Lectures on Anomalies and SPT Phases

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Abstract

This is a note on the lectures given by Prof. Yuji Tachikawa in TASI 2019. Focusing on viewing anomalous QFT as living on boundaries of SPT phases.

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1 Overview

People has a long investigation on the topic of Anomalies in QFT.

In this note we try to give out some well known examples of anomalies and try to rewrite them in the language of SPT phases.

2 Example 1: Dirac Quantization

2.1 Anomalous Partition Function

2.1.1 Basic Setup

In many systems couple to a $U(1)$ gauge field

- Particle confined in S^1
- Dirac Fermion ...

These model arises with a partition function:

$$Z(S^1, g) = \exp \left(i q \oint_C dx A \right) \quad (1)$$

For a mathematical defined $U(1)$ gauge field, we admit gauge transformation both small and large. For a suitable normalization, we can get

- Small gauge transformation: $A \rightarrow A + d\lambda$
- Large gauge transformation: for a close loop $\int_C dx A \rightarrow \int_C dx A + 2\pi n$

Remark

In EM, this large gauge transformation may exist due to physically the AB phase and to have non globally well defined gauge field. Moreover, the integral of curvature over a closed surface is quantized, which is physically the Dirac quantization condition of charge.

Here, we can just understand it as a mathematical property of $U(1)$ connection, as well as the fact that the integral of curvature over a closed surface is quantized. And we assume that EM field can be described by a $U(1)$ gauge field.

Thus under large gauge transformation, the partition function transforms as:

$$Z(S^1, g) \rightarrow Z(S^1, g) \exp(2\pi i q n) \quad (2)$$

if the coupling q is not an integer, the partition function is not invariant under large gauge transformation, which is an anomaly.

Bibliography